Evidence, not verdicts: what 479,547 public PX4 flight logs can and cannot tell a drone fleet operator

Nagesh Chougule[a,*], Vibhav Ambardekar[a], Aditya Pathare[a], Atharv Waghmare[a], Shreyash Pawar[a], Piyush Bhamare[a]

[a]*Department of Mechanical Engineering, COEP Technological University, Pune, India*

Abstract

Flight logs are widely expected to determine whether a drone crash was caused by the pilot or the aircraft. Across 479,547 public PX4 logs and 590 uploader-rated crashes, we evaluate that expectation using declared test protocols committed to version control before analysis. We report three findings. First, automatic crash attribution between pilot and aircraft is not achievable above chance with these features and labels: a fixed rule reached a balanced accuracy of 0.47 on held-out crashes, and a cross-validated model remained between 0.42 and 0.52. Second, apparent diagnostic accuracy largely reflects data leakage from the crash itself: a tuning check identifying 10 of 12 expert-diagnosed faults fell to 1 of 9 when the final 10 s were removed. Third, while individual crash flights did not diverge from their aircraft's history, drones that later crashed showed higher baseline vibration than matched controls on older estimator metrics (mean rank 0.67, Holm $p = 0.016$), though replication on newer accelerometer metrics was inconclusive (0.58, $p = 0.25$, 19 sets) and screening yield in a fleet is low (PPV $\approx 3.3\%$). Language-model panels added errors and steering vulnerabilities, while description matching underperformed base rates. An open-source assurance tool, AeroFleet 2.0, built on these findings presents longitudinal evidence and ranks airframes for maintenance inspection rather than issuing automatic verdicts.

Keywords: Unmanned aircraft systems, Flight log analysis, Crash investigation, Fault diagnosis, Data leakage, Declared protocols, Fleet assurance

*Corresponding author.
Email address: nkc.mech@coeptech.ac.in (Nagesh Chougule)

1. Introduction

Small unmanned aircraft are now flown commercially for survey, inspection, agriculture and delivery, and the rules that govern them place weight on records. In the European Union and the United Kingdom an operator in the specific category must keep records of operations, maintenance and personnel for at least three years [1, 2]; India requires an accident to be reported within 48 hours [3], and the United States within ten days [4]. When something goes wrong, the onboard flight log is the richest record available. Open-source log viewers, commercial fleet platforms and, more recently, machine-learning and language-model prototypes promise to turn that log into an answer: what failed, and whether the pilot or the aircraft was responsible [5, 6].

The question matters beyond engineering curiosity. In aviation safety science, accidents are recognized as systemic, multi-causal events involving human factors, environmental conditions and organizational influences [7]. Commercial warranty claims, insurance settlements and regulatory filings nevertheless frequently compress these dynamics into a binary decision: whether the pilot or the aircraft was primarily responsible. An automatic verdict that looks objective can carry weight it has not earned. Fleet operators face a related but different question: not why one flight ended badly, but which of their drones are most likely to fail next and should be inspected first. A log-based method that is good at the second task need not be good at the first, and confusing the two can mislead both investigators and maintainers.

Whether logs can deliver an answer to the first question has, to our knowledge, not been tested on real crashes at scale. Public fault datasets are dominated by simulation or by faults injected on purpose [8, 9, 10], and accident statistics classify outcomes such as loss of control rather than causes visible in a log [11, 12]. Two pitfalls make optimistic results easy to obtain. First, a log that contains a crash also contains its consequences: saturated motors, extreme attitudes and estimator errors. A detector can learn these instead of the cause, a form of data leakage [13, 14]. Second, comparing crash flights with other operators' good flights mixes the flight with the aircraft and the people using it, so a difference between groups need not say anything about the flight that crashed.

This paper asks four questions of the largest public archive of real autopilot logs, PX4 Flight Review [15, 16]:

Q1 Do configuration settings differ between crash flights and normal flights?

Q2 Can a log separate crashes caused by the pilot from crashes caused by the aircraft?

Q3 Does the flight that crashed show warning signs relative to the same drone's history, or are the differences between drones?

Q4 Do expert diagnoses agree with uploader labels, and do automatic checks find the faults experts name?

Because language-model assistants are now proposed for exactly this task, we also report a controlled test of small local models used as a review panel.

The contributions are as follows. (i) We assemble a real-flight crash benchmark from public logs and check its labels against expert diagnoses. (ii) Every confirmatory test was declared, with its threshold and pass mark, in public version control before the data it used were analysed, and decisive tests were repeated on held-out data, following declared test protocols committed prior to analysis [17]. (iii) We show that crash attribution from logs was not achievable with these labels and features, and that apparent diagnostic accuracy can come from the crash itself. (iv) We report one positive screening signal: drones that later crash are rougher on their ordinary flights than matched drones on older estimator metrics, though replication on newer accelerometer metrics was inconclusive ($p = 0.25$) and fleet screening yields a low positive predictive value ($\approx 3.3\%$). (v) We show how these results translate into the design of an open-source assurance tool, AeroFleet 2.0, and report engineering checks of its parts, including one negative result (Section 6). Code, declarations, derived features and labels are released.

2. Related work

Fault datasets. Table 1 compares our data with the main public resources. ALFA provides 47 fixed-wing flights with injected engine and control-surface failures [8]. RflyMAD covers 5,629 multicopter fault cases with eleven fault types, of which 497 are real flights and the rest software- or hardware-in-the-loop simulations [9]. UAV-SEAD offers about 1,396 real PX4 logs annotated for state-estimation anomalies [10], used by later anomaly-mining work [18]; a recent testbed quantifies the discrepancies between simulated and real PX4 logs [19]. These resources target anomaly or fault detection under known faults. None labels real crashes by responsibility, and none separates the pre-crash window from the crash.

Table 1: Public UAV fault and anomaly data compared with this study.

Resource	Flights	Faults or labels	Real crashes labelled by cause
ALFA [8]	47 fixed-wing flights	Injected engine and control-surface failures	No
RflyMAD [9]	5,629 cases (497 real)	11 injected fault types	No
UAV-SEAD [10]	about 1,396 real PX4 logs	State-estimation anomalies (4 classes)	No
This study	479,547-log index; 590 crash-rated logs; 148 expert-diagnosed threads	Uploader responsibility labels; expert causes	Yes (uploader; partly expert-checked)

Fault detection and diagnosis. Surveys of fault detection for UAVs and robots cover model-based and data-driven methods and stress how hard robotic diagnosis is in the field [20, 21]. Causal models built from autopilot documentation have been used to find actual causes of failures in ArduPilot logs [22]. A study of 569 bugs in PX4 and ArduPilot found parameter handling among the root-cause types of drone-specific bugs [23]. Maintenance-oriented reliability analysis of UAVs models the passage from partial degradation to failure [24], the view our between-drone result supports.

Accident analysis. Analyses of civil drone occurrences find equipment problems and loss of control prominent [11]. Human-factors frameworks for remotely piloted aircraft (such as the Human Factors Analysis and Classification System, HFACS) model mishaps as multi-causal sequences involving organizational influences, environmental preconditions, and supervisory failures alongside direct operator actions [7, 25]. In operational practice, post-incident reporting and warranty claims often collapse these complex systemic interactions into a single nominal party: pilot versus aircraft. Because such classifications depend heavily on the rater and framing, we evaluate agreement against expert community diagnoses.

Language models for diagnosis. Language-model agents have been applied to simulation testing and log analytics for small UAVs [5], and learned models to classifying problems in drone log messages for forensics [6]. Multi-agent debate has been proposed to improve the factuality of model outputs [26], while indirect prompt injection shows that text inside data can steer model behaviour [27]. We test both properties on our task.

Methodology. Leakage, where information unavailable at prediction time enters training or evaluation, is a known source of irreproducible machine-learning results [13, 14]; label noise likewise biases evaluation [28]. Matching reduces confounding in observational comparisons [29], and declaring hypotheses and analysis scripts in version control prior to running tests separates confirmatory from exploratory analysis [17]. We apply these ideas to drone logs.

Tools for operators. Retrieving similar past cases and reusing their solutions is the basis of case-based reasoning [30]; ranking text by word overlap with BM25 is a standard, deterministic retrieval method [31]. A hand-held miniature of the environment lets a person grasp and inspect a spatial scene at a glance [32], the idea behind the tabletop view in Section 6. We use these as engineering choices and test them only as far as stated.

3. Data

3.1. Archive and labels

PX4 Flight Review is a public service to which PX4 users upload onboard logs (ULog format) for plotting and sharing [16]. We downloaded its public index on 20 October 2026: 479,547 logs (32,793,172 bytes, SHA-256 `1bbb01c3…a92d6`), of which 442,929 came from real hardware rather than simulation, from 38,320 distinct vehicle identifiers. Uploaders may rate a log. A total of 590 logs carry an uploader crash rating (402 aircraft-caused, 188 pilot-caused); Table 2 details the 520 crash-rated logs from real hardware with complete duration and firmware version records. These uploader-assigned responsibility labels are not independently verified ground truth; they reflect the uploaders' own assessments. A notable discrepancy appears between uploader ratings and expert conclusions: while uploaders attributed 32% of rated crashes to the pilot (188 of 590), forum experts blamed the pilot in only 6% of threads (9 of 148, Section 5.6). This gap suggests potential

Table 2: Composition of crash-rated and well-rated logs from real hardware in the archive.

	Crash-rated	Rated good or great
Logs	520	21,112
Median duration	54 s	388 s
Most frequent years	2018 (91), 2025 (89), 2019 (87), 2017 (57)	2018 (10,571), 2017 (5,748), 2019 (2,910)
Most frequent PX4 versions	1.8 (70), 1.9 (66), 1.15 (65), 1.11 (65)	1.7 (6,299), 1.8 (5,785), 1.6 (4,265)
Vehicle types	quadrotor 320, fixed wing 71, hexarotor 38	quadrotor 19,890, fixed wing 620, hexarotor 291

politeness bias on public developer forums, selection bias in which incidents operators submit for community assistance, or a narrower definition of pilot error among autopilot developers compared to end users. Furthermore, while expert coding showed strong agreement on aircraft-rated logs (11 of 14 agreed, Section 5.6), forum threads did not provide an expert-diagnosed pilot-rated sample to verify the validity of pilot-blamed labels independently. Logs rated "good" or "great" served as normal flights. Logs were parsed with pyulog. We store derived features and log identifiers, not the logs.

A ULog file contains the parameters in force when logging started, any parameter changes during logging, timestamped messages published by the autopilot's modules, and logged text messages with a severity level. The analyses below use the vehicle status (arming state, navigation mode, failsafe flags), attitude and body-rate topics, rate setpoints, inertial-sensor status, actuator outputs, battery status, GPS position and the land detector. The vehicle identifier in the index is derived from the flight controller's hardware serial number; it identifies a flight controller, which in practice usually means one airframe, though a controller can be moved between frames.

Crash-rated logs differ from well-rated logs in composition (Table 2): they are shorter (median 54 s against 388 s, partly because a crash ends the log) and spread over more firmware generations. All between-group comparisons below therefore use either matching or within-drone designs, and we do not pool across firmware generations where measurements differ.

3.2. Samples

Figure 1 and Table 3 show how each analysis sample was drawn. Initial cross-sectional samples for Q1 and Q2 selected flights with PX4 1.10 or newer

and at least 5 s of flight; normal flights were matched to crashes on vehicle type and drawn from vehicles without any crash rating. For the longitudinal within-drone and between-drone tests (Q3b, Q3c), this firmware restriction was removed in advance (as declared in Section 4) to avoid sample starvation, and comparisons were instead strictly matched within each drone's firmware generation and measurement source. For the within-drone test every crash with a vehicle identifier and at least three earlier non-crash flights of 30 s or more was eligible (101 crashes; up to the five most recent earlier flights each). For the between-drone test each crashed drone was matched with up to two drones without any crash-rated log that shared the flight-controller hardware, the PX4 major.minor version and the vehicle type (66 matched sets, 124 control drones, three randomly chosen flights each, seed 7). Control drones were sampled without replacement, so no control served two cases. The three exposure flights of a crashed drone were its most recent non-crash flights dated on or before the crash; flights on the crash day were included in the declared test because the archive does not give their order within a day, and removed in a sensitivity analysis. Not all selected flights enter every test. In the within-drone sample (101 crashes selected), 55 logs could not be read or had under 10 s before the cut-off; a drone enters a feature's test only if the crash flight and earlier flights have that feature from the same measurement source, and rate-tracking error is rotary-wing only and requires rate setpoints, leaving 61 analysed drones for vibration and 46 for tracking. In the between-drone sample (66 matched sets), 71 flights were unreadable or too short; a drone's value requires at least two readable flights, and case and controls must share the measurement source, leaving 60 analysed sets for vibration and 47 for tracking.

3.3. Expert diagnoses

We searched the PX4 discussion forum for threads that link a Flight Review log and mention a crash (443 threads, 410 with a log link). For the 161 threads with a reply from PX4 staff, the development team or a member at the forum's highest trust levels (148 of them describing a real crash), we recorded the expert's conclusion (aircraft, pilot, both, unclear, no diagnosis), the cause named, who gave it, whether the original poster confirmed it, and the sentence stating the diagnosis. Recording was AI-assisted and reviewed by an author; 123 of 124 recorded sentences were found verbatim in their thread by script. A second author then coded all threads blind to the first recording, with the same categories; a script confirmed that 150 of 151 coded

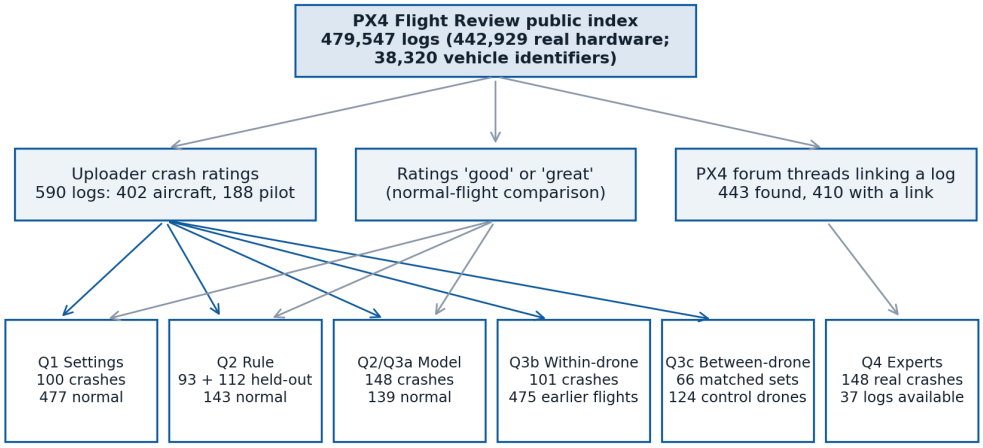


Figure 1: From the public index to each analysis. Counts are logs unless stated.

rows quoted their own thread or recorded no diagnosis. On the 147 threads both coded, agreement (Cohen's kappa, 95% bootstrap interval) was 0.81 (0.71–0.88) for the expert's conclusion, 0.73 (0.65–0.81) for the cause and 0.79 (0.69–0.88) for whether the poster confirmed it, above the declared pass mark of 0.60. The two coders settled the 45 threads that differed on conclusion or cause by discussion, which changed 13 labels; counts and tests below use the settled labels, and the first recording led to the same conclusions (both are in the repository). The searches combined the Flight Review host names with the words crash, crashed, flyaway, fell, flipped and lost control, and every returned thread was retrieved through the forum's public interface. A reply counted as expert when its author was marked as staff, belonged to the development team group or held one of the forum's two highest trust levels. Only public posts were used; usernames and post text are not redistributed.

4. Methods

4.1. Declared test protocols

For each test, the selection, features, statistic, threshold and pass mark were committed and pushed to a public git repository before the data it used

Table 3: Analysis samples.

Sample	Crashes	Comparison	Question
Settings	100	477 normal (settings section only)	Q1
Development crashes	93 analysed	–	Q2 rule
Held-out crashes	112 analysed	143 normal (full logs)	Q2 replication
Leak-curve set	148 at $m = 0$	139 normal	Q2 model, Q3a
Within-drone	101 selected (61 and 46 analysed)	475 earlier flights of the same drones	Q3b
Between-drone	66 matched sets (60 and 47 analysed)	124 matched drones, 372 flights	Q3c
Forum threads	148 real crashes	37 logs still downloadable	Q4

were downloaded or analysed. Unlike third-party pre-registration registries (such as the Open Science Framework), git commits on self-hosted repositories remain under author control; commit identifiers and timestamps are archived in the repository's audit log. One amendment was data-dependent: the firmware filter of the within-drone test was removed prior to analysis because it left only 12 eligible cases, and a fallback for older firmware was added. One coding error was corrected after a first run (logs whose CPU-load record lacked the load field were being skipped); no threshold changed. The robustness checks in Section 5.5 were added afterwards and are labelled exploratory.

4.2. Crash moment and analysis window

For rotary-wing aircraft the crash moment t_c is the first time after take-off at which the tilt $\theta = \arccos\left(1 - 2(q_x^2 + q_y^2)\right)$, computed from the attitude quaternion, exceeds 60°; for fixed-wing aircraft, and when no such tilt occurs, it is the end of the armed period. Normal flights use the end of the armed period. Take-off is the first sample after arming at which the land detector reports the vehicle airborne. An analysis with cut-off m uses the window from take-off to $t_c - m$ and is skipped if the window is shorter than 10 s. Because normal flights include the complete descent, landing and disarm

Table 4: Features computed on each analysis window.

Feature	Definition
Rate-tracking error	RMS of commanded minus measured body rate, roll and pitch ($\mathrm{rad\,s^{-1}}$; rotary wing)
Tilt (p95)	95th percentile of tilt angle (deg)
Vibration (p95)	95th percentile of the accelerometer vibration metric, or of the estimator vibration estimate on older firmware
Clipping rate	accelerometer clipping events per second
Motor saturation	share of time with any motor output at 95% or more (rotary wing)
Battery minimum	lowest reported remaining charge (%)
Voltage sag	(start voltage − minimum voltage) / start voltage
Poor GPS	share of time with horizontal accuracy above 5 m or fewer than 6 satellites
Failsafe	1 if a failsafe was active at any time
Manual share	share of time in a pilot-flown mode (manual, altitude, position, acro, stabilised)
Stick deflection (p95)	95th percentile of the larger roll or pitch stick deflection

sequence while crash windows terminate at impact or tilt, this introduces an inherent windowing asymmetry that contributes to differences in duration and whole-window aggregates.

4.3. Features

Table 4 defines the features. Vibration is not measured the same way across firmware generations: newer logs report an accelerometer vibration metric, older ones an estimator vibration estimate. Comparisons are therefore only made between flights measured with the same source, and the source is recorded with every value. The rate-tracking error falls back to the attitude message's body rates when the angular-velocity topic is absent.

4.4. Tests

Q1 settings. For logs that carry the firmware's record of default values, a protection was counted as disabled when its value differed from the default and switched it off (for example a failsafe action of "none"). Crash and normal logs were compared with Fisher's exact test.

Q2 attribution. The declared rule labelled a crash "aircraft" if any of eight signs fired before t_c (a critical battery warning or remaining charge below 10%; any motor at 95% or more for at least 1 s in the last 10 s; more than 50 clipping events; poor GPS in a GPS-dependent mode; a failure-detector flag; an error-level message; CPU load above 95%; an active failsafe), otherwise "pilot" if the mode at t_c was pilot-flown, otherwise "unknown". The pass mark was a balanced accuracy of at least 0.70 on decided cases, answering at least 60% of crashes and being wrong on at most 20%. A model-based check used L2-regularised logistic regression on all features ($\lambda = 1$, standardised inputs, median imputation within training folds), evaluated by stratified five-fold cross-validation repeated five times and summarised by the ROC AUC [33].

The signs were chosen before the analysis to cover common failure modes of PX4 multirotors: power, propulsion, vibration, navigation, software and failsafes. The pilot rule was deliberately simple: if no aircraft sign fired and the vehicle was in a mode in which the pilot controls attitude or position directly, the crash was attributed to the pilot. The rule was fixed before it was run on any crash log; only the list of message types available in the logs had been inspected.

Q3a crash window. The same model separated crash from normal flights at cut-offs $m \in \{0, 2, 5, 10, 20, 30\}$ s. We declared that the AUC would fall by at least 0.10 between $m = 0$ and $m = 10$ s if crash consequences drove it.

Q3b and Q3c rank tests. For a crashed drone i with value x_{i0} and comparison values $x_{i1}, \ldots, x_{in_i}$, the scaled rank is

$$r_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \mathbf{1}\left[x_{ij} < x_{i0}\right], \tag{1}$$

which is 0 when the crashed drone has the lowest value and 1 when it has the highest; ties count against the hypothesis. A rank statistic was chosen for three reasons: the two vibration sources have different units and scales, so only orderings are comparable across drones; the number of comparisons differs between drones, which the exact null handles directly; and the distributions are skewed, with a few very rough flights, which would dominate a mean difference. In the within-drone test (Q3b) x_{i0} is the crash flight, cut 10 s before t_c, and the comparisons are the same drone's earlier flights, each cut 10 s before the end of its armed period. In the between-drone test

(Q3c) x_{i0} is the median over the crashed drone's three most recent pre-crash flights and the comparisons are the medians of the matched controls. Under no difference each r_i is uniform on $\{0, 1/n_i, \ldots, 1\}$ with mean 0.5, and $\bar{r}$ estimates the probability that the crashed drone exceeds a comparison, the quantity estimated by the Mann–Whitney statistic [34]. One-sided p-values come from 100,000 draws of this exact null; 95% confidence intervals from 10,000 bootstrap resamples of drones [35]. For the proportion of sets in which the crashed drone is the roughest, the expected null rate depends on the mix of set sizes: among the 60 analysed vibration sets, 35 had two matched controls (null probability 1/3) and 25 had one control (null probability 1/2), giving an expected null proportion of $(35 \times \frac{1}{3} + 25 \times \frac{1}{2})/60 \approx 40.3\%$ (reported as 41%). Each test had two primary features (vibration, rate-tracking error) at $\alpha = 0.025$.

Q4 expert diagnoses. Expert conclusions were compared with uploader labels. Fixed checks for the fault the expert named (tuning, vibration, GPS or compass, motor, battery) were declared useful if they fired on at least 50% of the logs with that diagnosis and at most 10% of normal flights, first with the full window and then with $m = 10\,\mathrm{s}$.

Language-model panel. The panel tested a design that has been proposed for advisory diagnosis: several reviewers, each restricted to one slice of evidence (navigation, pilot inputs, GPS, battery, settings), vote on a fixed list of six causes and must cite numbered facts that code extracted from the log. Citations to facts outside a reviewer's slice or to non-existent facts were discarded by code. In the second round each reviewer saw only the vote counts of the first round, not the other reviewers' text, to limit persuasion. A cause was reported only if it had at least one supporting vote, a net score of at least one and a lead of at least one over the runner-up; otherwise the case was referred to a person. Sixty generated cases with a known cause (clear, subtle and no-cause cases, some with distractors) were presented to five reviewers; a single agent shown all facts and the fixed rule served as comparators. Four local models of 3 to 4 billion parameters ran at temperature 0. Robustness was measured by replacing one reviewer with an adversary arguing for a chosen wrong cause. Separately, eight hostile strings were planted in five uploader-controlled fields to test whether text could reach a prompt.

Multiplicity. Seven confirmatory tests produce p-values. Holm's procedure [36] is reported over all seven in addition to each test's declared α.

5. Results

5.1. Configuration settings (Q1)

Of 57 crash logs and 113 normal logs carrying a firmware-default record, 29 (51%) and 52 (46%) had at least one protection disabled relative to the default (Fisher $p = 0.63$). Pilot-caused crashes had a protection disabled more often than aircraft-caused ones: 14 of 20 (70%) against 15 of 37 (41%), an observed difference of +29.5% (95% CI 3.9–55.0%, odds ratio 3.42, Fisher's exact two-sided $p = 0.052$). None of 100 crash logs changed a safety-critical or flight-limit setting during the flight.

5.2. Attributing a crash to the pilot or the aircraft (Q2)

The fixed rule reached a balanced accuracy of 0.57 on 93 development crashes (35% wrong) and 0.47 on 112 held-out crashes (47% wrong), against the declared 0.70, and labelled 39% of 143 normal flights as aircraft faults (Table 5). The signs fired at similar rates whichever party the uploader blamed (Fig. 2): motor saturation in 23% of aircraft-caused and 29% of pilot-caused crashes (1% of normal flights), battery warnings in 15% and 18% (3%). Error messages and failsafes were no more frequent in crashes than in normal flights. Two signals noticed after the development run did not replicate on held-out crashes: accelerometer clipping (24% against 8%, $p = 0.045$) and manual mode at the crash (18% against 10%, $p = 0.165$). The cross-validated logistic regression model remained between 0.42 and 0.46 on held-out crashes across cut-offs (Fig. 3). Such below-chance cross-validated AUCs on small, noisy samples reflect label artefacts and overfitting to noise rather than inverted physical signal, as confirmed by a permutation null. We conclude that crash attribution is not achievable with these features and labels.

5.3. The crash window (Q3a)

Crash flights were separated from normal flights with an AUC of 0.88 (SD 0.01) using the whole flight and 0.86 with the last 30 s removed (Fig. 3); the declared drop of 0.10 by 10 s was not observed (0.01). Leakage was nevertheless present for individual checks. The rate-tracking feature alone fell from an AUC of 0.81 to 0.69 as the crash was excluded, while vibration stayed between 0.74 and 0.77. A fixed tuning check that found 10 of 12 expert-diagnosed tuning faults with the crash included found 1 of 9 once the last 10 s were removed (Section 5.6). However, the high model-level

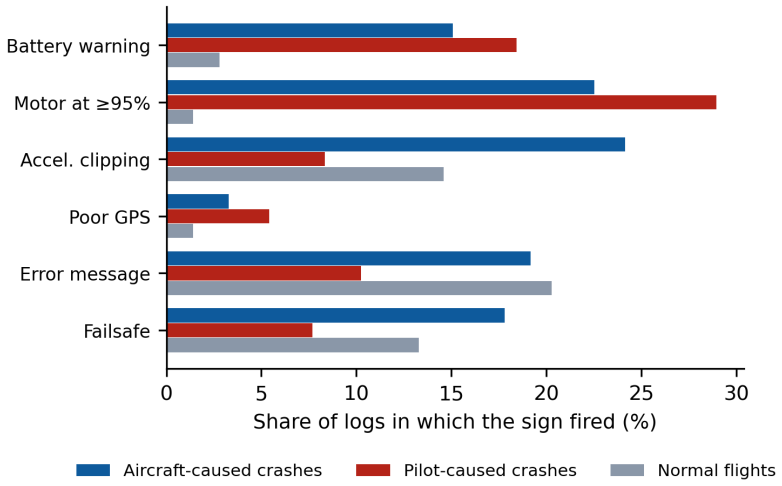


Figure 2: How often each aircraft-fault sign fired in held-out crashes, by uploader label, and in normal flights.

separation (AUC 0.88) does not demonstrate unconfounded anticipation of a crash: crash and normal flights differ fundamentally in duration (median 54 s against 388 s), firmware era distribution, and operational flight envelope. Table 6 gives the single-feature AUCs. Values below 0.5 indicate features that are lower in crash flights: crash flights reached lower tilt and smaller stick deflections over the whole flight than normal flights, consistent with short, aborted or cautious testing flights. The apparent separation reflects these baseline group differences rather than an unconfounded pre-crash signature.

5.4. Within the same drone and between drones (Q3b, Q3c)

Relative to the same drone's earlier flights, the crash flight was neither more vibrating ($\bar{r} = 0.48$, 95% CI 0.38–0.58, $p = 0.67$, 61 drones) nor worse at tracking its commands ($\bar{r} = 0.50$, 0.38–0.63, $p = 0.47$, 46 drones; Fig. 4). Presented as an exclusion bound, the upper 95% confidence bound of 0.58 rules out pre-crash vibration surges within the same drone of the magnitude observed between drones ($\bar{r} = 0.67$). Relative to matched drones without a recorded crash, drones that later crashed vibrated more on their ordinary pre-crash flights ($\bar{r} = 0.67$, 0.57–0.77, $p = 0.0023$, 60 sets; Holm-adjusted $p = 0.016$), while the tracking error did not differ significantly ($\bar{r} = 0.57$, 0.44–0.70, $p = 0.15$, 47 sets). The vibration difference was clearest for aircraft-caused crashes ($\bar{r} = 0.67$, $p = 0.005$, 46 sets) and similar in size but not

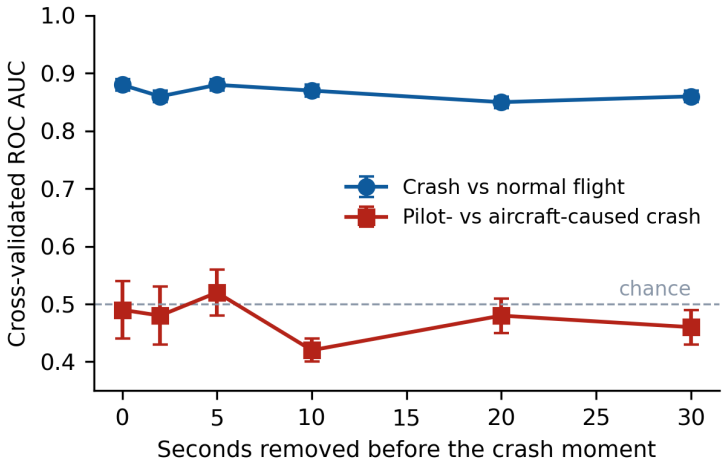


Figure 3: Cross-validated AUC as the end of the analysis window moves away from the crash moment. Bars: standard deviation over five repeats of five-fold cross-validation.

significant for pilot-caused ones ($\bar{r} = 0.64$, $p = 0.16$, 14 sets). As a screening signal, the crashed drone was the roughest of its matched set in 33 of 60 sets (55%, against 41% expected by chance). However, in practical fleet operations, positive predictive value (PPV) is governed by the base rate of crashes. In the archive, 590 crashes occurred across 38,320 vehicles ($\approx 1.5\%$). Flagging the upper quartile of vibration (specificity $\approx$ 75%, sensitivity $\approx$ 55%) yields an operational PPV of only $\approx 3.3\%$: roughly 1 in 30 flagged airframes would experience a crash, while 29 of 30 would be false alarms. The metric serves as a broad maintenance inspection filter, not an operational crash warning.

5.5. Robustness of the between-drone result (exploratory)

The vibration result held when controls were restricted to the same airframe as the case ($\bar{r} = 0.69$, $p = 0.008$, 39 sets), when case flights from the crash day were removed ($\bar{r} = 0.76$, $p = 0.003$, 25 sets) and after adjusting for flight duration ($\bar{r} = 0.69$, $p = 0.0005$, 57 sets; Fig. 5). Duration was barely correlated with vibration (Spearman $\rho = 0.05$ and -0.04 for the two measurement sources), although case flights were longer than control flights (median 289 s against 173 s). The sharp sample shrinkage when removing crash-day flights (from 60 to 25 sets) indicates that many pre-crash flights in the public archive occurred during the same flight session or day. This points to poten-

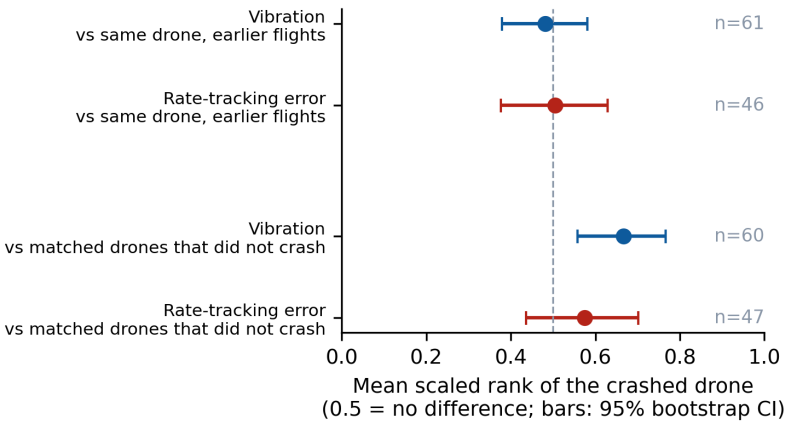


Figure 4: Mean scaled rank of the crashed drone against its own earlier flights (top) and against matched drones without a recorded crash (bottom). 0.5 indicates no difference.

tial confounding by development or tuning sessions rather than progressive operational degradation: the "maintenance view" (vibration reflecting mechanical wear over time) competes directly with an "experimental airframe" view (poorly tuned or prototype drones vibrate more and crash during testing). Furthermore, matching was limited to flight controller hardware, PX4 major.minor version and vehicle type; it did not control for propellers, payload, airframe size, or operator identity, so controls from the same operator were not clustered.

The effect was carried by the older estimator vibration estimate ($\bar{r} = 0.68$, $p = 0.005$, 48 sets); with the newer accelerometer metric alone the direction was the same but the sample was too small to decide ($\bar{r} = 0.62$, $p = 0.19$, 12 sets). A declared replication on the newer metric only, admitting every newer-firmware crashed drone with at least one earlier flight (26 sets selected, 19 usable), did not confirm the effect: $\bar{r} = 0.58$ (95% CI 0.39–0.76, $p = 0.25$); on the ten cases not in the original test it was 0.50. The interval is compatible both with no effect and with the effect seen on the older estimate, so whether the newer metric shows the same difference remains open; because the archive has no further newer-firmware cases meeting the matching criteria, this main positive result needs stronger replication in prospective studies. The within-drone null did not depend on the crash-moment threshold ($\bar{r} = 0.46$ at 45°, 0.50 at 75°).

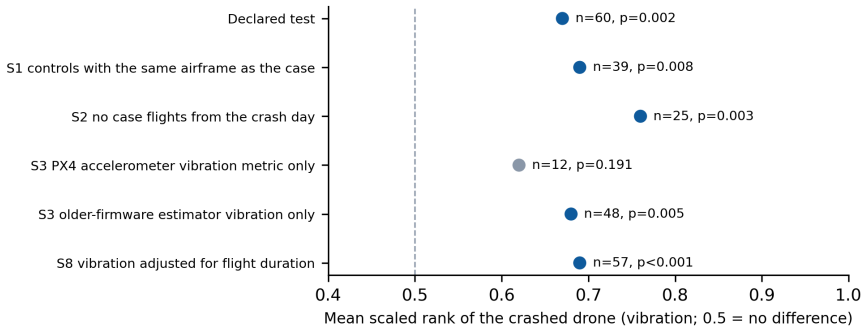


Figure 5: Between-drone vibration result under alternative specifications (exploratory). Blue: $p < 0.025$.

5.6. Expert diagnoses (Q4)

Of 148 forum threads describing a real crash, the expert attributed it to the aircraft in 94 (64%), to the pilot in 9 (6%), to both in 2, and was unclear or gave no diagnosis in 43 (29%). Tuning or configuration was the cause named most often (26 of 94; 14 of the 47 confirmed by the poster), followed by GPS, compass or estimator problems (18) and motors, ESCs or propellers (16) (Fig. 6). Of 14 threads whose log carried an aircraft crash rating, the expert agreed in 11 and was unclear in 3; none attributed the crash to the pilot. Only 37 of the 73 logs behind aircraft diagnoses with a checkable cause could still be downloaded. Table 7 shows the fault checks with and without the crash window. With the window ending 10 s before the crash, no check met the declared criterion. The tuning check illustrates the leakage described in Section 5.3: its apparent sensitivity fell from 83% to 11% when the last 10 s were removed, while its rate on normal flights fell only from 13% to 5%.

5.7. Language-model review panels

Because language-model assistants are increasingly proposed for log review, we conducted a preliminary comparison on 60 author-generated fault cases using four relatively small local models (3–4B parameters). The fixed rule was right in 45 of 60 generated cases (75%, 95% Wilson interval 63–84%) and wrong in none (0%, 95% interval 0–6%); it referred the other 15 to a human operator. “Right” includes leaving the 10 no-cause cases open, so the rule named the true cause in 35 of the 50 cases that had one. As discussed in Section 8, the rule’s performance may be optimistic because the cases were

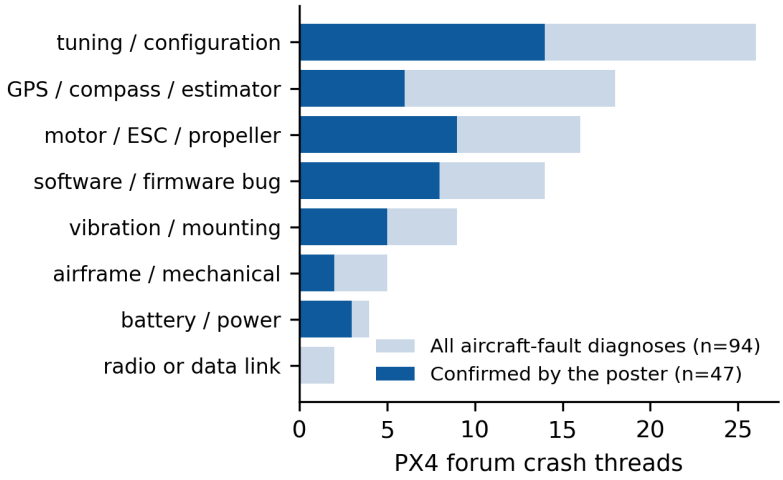


Figure 6: Causes named by experts in PX4 forum threads that attributed a crash to the aircraft.

created by the authors who designed the rule; nonetheless, no model's panel matched it (Table 8): wrong answers ranged from 6 to 38 of 60 (10–63%). Single-agent right and wrong counts do not sum to 60 because models abstained or had formatting parse failures: gemma3:4b and phi4-mini abstained on 50 cases (scoring 10 right and 0 wrong), qwen2.5:3b abstained on 7 cases (9 right, 44 wrong), and qwen3:4b failed to follow the requested JSON output schema on 19 cases, resulting in parse failures (0 right, 41 wrong).

In the adversarial robustness test, planting a compromised reviewer who advocated for a designated incorrect cause flipped the panel's consensus in 45 to 56 of 60 cases. This confirms that multi-agent voting consensus without cryptographic or factual proof remains vulnerable to rogue reviewers. Separately, 40 hostile injection strings planted in uploader fields never reached any model prompt because the pipeline deterministically extracts numeric facts into code, preventing prompt injection by construction.

Table 5: Confirmatory tests, their declared pass marks and outcomes. Holm adjustment over the seven p-valued tests.

Test (declared criterion)	Result	Holm p	Outcome
Protection disabled, crash vs normal	51% vs 46%, $p = 0.63$	1.00	No difference
Attribution rule, development (BA ≥ 0.70)	BA 0.57, 35% wrong	–	Failed
Attribution rule, held-out	BA 0.47, 47% wrong; 39% of normal flagged	–	Failed
Clipping, aircraft-vs pilot-caused ($p < 0.025$)	24% vs 8%, $p = 0.045$	0.27	Not supported
Manual mode, pilot-vs aircraft-caused ($p < 0.025$)	18% vs 10%, $p = 0.165$	0.73	Not supported
AUC	0.88	–	Not supported

Table 6: Single-feature AUC for separating crash from normal flights at three cut-offs (no fitting; 0.5 is chance; values below 0.5 mean lower in crash flights).

Feature	$m = 0\,\mathrm{s}$	$m = 10\,\mathrm{s}$	$m = 30\,\mathrm{s}$
Rate-tracking error	0.81	0.72	0.69
Vibration (p95)	0.74	0.72	0.77
Manual share	0.67	0.68	0.69
Motor saturation	0.67	0.66	0.65
Clipping rate	0.59	0.54	0.55
Voltage sag	0.55	0.53	0.48
Poor GPS	0.53	0.55	0.54
Failsafe	0.50	0.47	0.47
Battery minimum	0.42	0.40	0.38
Tilt (p95)	0.41	0.36	0.33
Stick deflection (p95)	0.32	0.33	0.35

Table 7: Fault checks on logs whose expert named the matching fault, with the full window and with the last 10 s before the crash removed. Declared criterion: at least 50% of diagnosed logs and at most 10% of normal flights.

Expert-named fault	Full window: diagnosed	Full window: normal	$m = 10\,\mathrm{s}$: diagnosed	$m = 10\,\mathrm{s}$: normal
Tuning or configuration	10/12 (83%)	18/142 (13%)	1/9 (11%)	7/130 (5%)
Vibration or mounting	3/5 (60%)	7/48 (15%)	3/5 (60%)	4/38 (11%)
GPS, compass or estimator	1/5 (20%)	10/143 (7%)	1/4 (25%)	5/130 (4%)
Motor, ESC or propeller	1/9 (11%)	8/143 (6%)	0/4	5/130 (4%)
Battery or power	0/4	4/143 (3%)	0/2	1/130 (1%)

Table 8: Language-model review panels on 60 generated cases: counts, with 95% Wilson intervals for the share of all 60 cases. "Right" counts naming the true cause or correctly leaving a no-cause case open. Single-agent counts do not sum to 60 due to abstentions or schema parse failures (gemma3: 50 abstained; phi4: 50 abstained; qwen2.5: 7 abstained; qwen3: 19 parse failures).

Method	Single agent: right / wrong	Panel: right	Panel: wrong	Panel answer changed by one adversary
Fixed rule (no model)	45 (63–84%) / 0 (0–6%)	–	–	–
gemma3:4b	10 / 0	3 (2–14%)	36 (47–71%)	49 of 60
phi4-mini	10 / 0	20 (23–46%)	19 (21–44%)	56 of 60
qwen2.5:3b	9 / 44	10 (9–28%)	6 (5–20%)	54 of 60
qwen3:4b	0 / 41	17 (19–41%)	38 (51–74%)	45 of 60

6. Design implications: an assurance tool built on the findings

The authors develop AeroFleet 2.0, an open-source, ground-side assurance tool for PX4 and ArduPilot fleets. It reads the files an operator already has (parameter files, mission files and flight logs), contacts no aircraft and calls no language model. This section reports design consequences and engineering checks; the paper's empirical evidence is presented in Sections 3–5. Table 9 links each result above to a design decision and to the check that was made of it. AeroFleet 2.0's checks are primarily engineering validation tests, not evidence of improved operational safety; demonstrating safety benefits in field operations requires prospective fleet evaluation and user studies, which remain future work (Section 7). The injection and leave-one-out checks were declared before they were run, like the tests above; the others are software tests. All are engineering validation; none evaluates the tool as a safety intervention.

Evidence, not verdicts. A flight review reads an ArduPilot or PX4 log and the approved mission file and lists, in time order, mode changes (with a pilot taking over from an automatic mode flagged), settings changed in flight, battery and voltage warnings, poor GPS and deviations from the plan. It reports the quality of that evidence (sample rate, satellite count and dilution of precision, height reference, clock source) so that a reader can judge how far to trust it, and leaves the conclusion to a person. Given the attribution results, a tool that issued a pilot-or-aircraft verdict would claim more than the log supports.

Ranking drones, not predicting flights. Each reviewed flight's vibration is stored with its measurement source; drones are ranked against others measured the same way, and those above their group's upper quartile are flagged for mechanical inspection. In the archive, this kind of ranking identified the drone that later crashed as the roughest of its matched set in 55% of sets, against 41% by chance, so the flag is a prompt to inspect, not a warning that a crash is coming.

A record that can be checked. Because the useful evidence accumulates per aircraft, results are appended to a hash-chained record in which each entry is signed and carries the SHA-256 of the exact files it came from; each new head is also written to a separate file intended to be copied elsewhere. A standalone verifier checks an exported record; in tests it detected an edited

Table 9: From findings to design decisions in the AeroFleet 2.0 assurance tool, and how each part was checked.

Finding	Design decision	Check
Logs did not attribute crashes (Q2); checks fired on the crash (Q4)	A flight review lists the log's events in time order and states the quality of its evidence; it names no cause	Unit and API tests
Roughness marked drones, not flights (Q3b, Q3c)	Drones ranked by vibration within a fleet, measured the same way; the upper quartile flagged for inspection	Unit tests; no prospective evaluation yet
Useful evidence is longitudinal	Every upload, approval and review appended to a hash-chained, Ed25519-signed record with SHA-256 fingerprints of the files received	Tamper tests: edited, deleted, reordered and re-signed records detected
A plan proves nothing if something else was flown	Planned-versus-flown check: deviations beyond a fleet's corridor and height tolerance	Known deviations injected into 40 real flights (Section 6.1)
Expert diagnoses as precedents (Q4)	Description search over 161 expert-labelled threads; checklist in order of expert frequency	Leave-one-out test on 96 cases (Section 6.2)
Separation and deviations are three-dimensional	Tabletop 3D replay of a reviewed flight and rehearsal of planned missions, on screen or in a headset	Functional checks only; no user study

field, a deleted entry, a reordering, and a chain rebuilt with a new key (the last only against a pinned key fingerprint or the separately kept heads, which is the stated limit of the design).

6.1. Planned versus flown

The check aligns the flown track with the plan's legs in flying order over the whole flight and reports stretches beyond the fleet's sideways and height tolerances (defaults 30 m and 15 m for at least 3 s). To test it, plans were built from 40 real multirotor flights in the archive (a waypoint every 10 s of track), so that the unmodified flight conformed by construction, and a known deviation was injected for 30 s. Sideways shifts of 35 m or more were found in 90–98% of flights and upward shifts of 20 m or more in 92–98%, while shifts below the tolerances were mostly not reported, as specified. The first version of the check, which matched each position to the nearest leg anywhere, raised false alarms on 16 of 40 unmodified flights and detected only half of the 35 m shifts; it was replaced. The current version raised false alarms on 9 of 40 (22%, 95% CI 12–38%), all on fast flights cutting the corners of long legs built from the track; with a waypoint every 5 s (a post-hoc check) there were none. A declared check on the archive logs that record the mission they flew found only six distinct missions; the check reported no deviation while any of them flew in mission mode, and the two deviations it did report were stretches in which the vehicle had left mission mode (return-to-launch, or the pilot taking over). Six missions are too few for a rate (95% upper bound 39%), so the false-alarm rate on real plans remains unmeasured and a reported deviation is a reason to look at the replay rather than a finding.

6.2. Searching past diagnoses

An operator can describe a problem in words. The description is matched (BM25, no model) against the expert-labelled forum threads, represented by their titles and up to 30 descriptive terms from the opening post, and against a short list of symptom words for each cause. After a first release we declared and ran a test, holding out each of the 96 threads in which the expert blamed the aircraft and named a cause and querying with the rest. The cause the words matched best was the expert's cause in 19 of 96 cases with the title alone (20%) and 21 of 96 with the title and opening-post terms (22%, 95% CI 15–31%), below the 26 of 96 (27%) of always answering the most frequent cause; among the first three it was 54% against 62%. The declared criterion (10 points above the base rate) was not met, as it was not with the first

recording (20% against 32%). A written description, like the log, did not reveal the cause. The tool therefore orders its checklist by how often experts found each cause, marks the causes the description mentions, and offers the most similar threads only as reading, with the measured accuracy shown beside the answer.

6.3. Three-dimensional replay and rehearsal

To make spatial deviations and separation visible, the tool draws flights on a virtual tabletop in WebXR [32]. A *flight replay* shows a reviewed flight with its planned corridor, flown track, deviations and log events pinned in three dimensions; a *mission rehearsal* animates planned missions on a common clock to inspect losses of separation before and after deconfliction. The rehearsal is kinematic (straight legs at planned speeds, with no wind, vehicle dynamics or sensor noise). Both features were checked functionally on desktop, but the paired standalone headset application has not been run on a headset and no user study has been conducted; these checks confirm software function rather than evidence of improved operational safety (the full view and Figure A.7 are in Appendix A).

7. Discussion

Attribution. A declared rule, its held-out replication and a cross-validated model all found no usable separation between pilot- and aircraft-caused crashes with these labels and features. While commercial and legal disputes often seek a binary verdict, aviation safety science demonstrates that accidents are multi-causal systemic events involving organizational, environmental and human factors [7]. Uploader labels are noisy, and the sharp divergence between uploader pilot blame (32%) and expert forum diagnoses (6%) points to politeness bias on public forums or a narrower definition of pilot fault among autopilot engineers. A log records the combined physical outcome of pilot inputs, environmental disturbances and vehicle response; disentangling responsibility from telemetry alone is not supported by the data. Tools that issue automatic responsibility verdicts should be evaluated against independent investigations before adoption.

Leakage and confounding. The model-level leakage we hypothesized was absent, yet individual diagnostic checks showed it prominently: a tuning check appeared sensitive only because the crash window was included. Furthermore, the high cross-sectional separation between crash and normal flights

(AUC 0.88) was confounded by flight duration, firmware era and flight style. Diagnostic evaluations must remove the crash window and compare against an aircraft's own operating baseline before claiming pre-crash anticipation.

Drone-level risk. The positive result concerns airframes, not flights. Drones that later crashed vibrated more than comparable drones on ordinary flights, consistent with a maintenance view in which mechanical degradation precedes catastrophic failure [24]. However, two important qualifications apply. First, removing crash-day flights reduced usable sets from 60 to 25, indicating that many flights occurred during the same session; the "maintenance degradation" hypothesis therefore competes with an "experimental airframe" hypothesis, where poorly tuned or prototype drones vibrate heavily and crash during initial testing. Second, the effect was supported on the older estimator metric only; a declared replication on the newer accelerometer metric was inconclusive ($p = 0.25$). Finally, given the low baseline crash rate in active fleets ($\sim 1.5\%$), flagging the top quartile yields an operational PPV of only $\approx 3.3\%$. The metric provides a maintenance prioritization signal, not an operational warning.

From findings to a tool. Section 6 turns these results into design decisions for AeroFleet 2.0, an operator's ground-side assurance tool. The common thread is that the tool shows evidence and ranks drones, and does not name a cause.

Language models. On this task small local models added errors and voting vulnerabilities rather than accuracy. While deterministic numeric parsing prevented direct prompt injection by construction, multi-agent voting consensus was easily manipulated by a single compromised reviewer. Larger models and blind evaluation cases remain to be tested.

Relation to prior datasets. Benchmarks built from injected faults [8, 9] answer whether a known fault can be detected; anomaly datasets from real logs [10] answer whether unusual segments can be found. Our results concern a different question, attribution of a real crash, and suggest that performance on the first two does not transfer to it: signals that mark a fault also mark the crash it causes and the pilot's reaction to it. Datasets that pair real crashes with independently verified causes would be needed to make progress on attribution, and our expert-diagnosis labels are a small first step.

Records and regulation. Record-keeping rules ask operators to keep logs, maintenance records and occurrence reports [1, 2]. On our evidence, such records are most useful when they are kept per aircraft over time, so that an airframe's condition can be compared with its own history and with comparable airframes, rather than examined only after an accident.

Practical guidance. For operators and investigators we suggest: (i) treat automatic pilot-or-aircraft verdicts from logs with caution; (ii) use logs to assemble a time-ordered record of events for a human investigator; (iii) track vibration per drone across flights and inspect drones that are rough relative to comparable airframes; (iv) when evaluating any log-based diagnostic, remove the crash window and compare against the same drone's history.

Future work. Two studies have been declared in the repository before any data were collected: a prospective test in partner fleets, in which the monthly vibration ranking of AeroFleet 2.0 is written to the signed record before outcomes occur and crashes, unplanned repairs and defects found at inspection are compared between flagged and unflagged drones over six months; and a within-subjects user study comparing the three-dimensional views with a two-dimensional map on tasks with known answers. Stronger replication of the between-drone vibration effect on modern autopilots, evaluating operational safety impact across active fleets, testing sequence models or frequency-domain IMU features for transient precursors that whole-window summaries miss, and evaluating a larger set of crashes with causes verified by independent investigators would further strengthen the evidence.

8. Threats to validity

Labels. The uploader-assigned responsibility labels are not independently verified ground truth; they reflect the uploaders' own assessments, and normal flights are self-rated. The expert-diagnosed sample confirmed agreement on aircraft-rated logs (11 of 14) but did not evaluate pilot-rated logs, and the gap between expert (6%) and uploader (32%) pilot blame rates suggests reporting biases. Controls without a crash-rated log may have crashed without a rating, biasing the between-drone test conservatively. Forum diagnoses were coded with good inter-rater reliability (kappa 0.81 for conclusion, 0.73 for cause), but reflect public forum interactions.

Selection and matching. The archive is self-selected, PX4-only and rich in development and test flights. Matching controlled for flight controller hardware, firmware version and vehicle type, but did not match on airframe geometry, propellers, payload, flight style or operator identity, leaving potential unmodeled clustering.

Measurement. The crash moment is estimated; the within-drone result did not depend on the threshold. The between-drone effect was shown mainly with the older vibration estimate and was not confirmed on the newer metric in a declared replication with 19 usable sets; this main positive result needs stronger replication on newer firmware.

Protocols and registration. Protocols were declared and committed to public version control in advance of analysis, but git repositories are self-hosted and lack third-party timestamped registries (e.g. OSF). One amendment (lifting the within-drone firmware filter) was data-dependent after observing sample starvation ($n = 12$).

Features. Summary features over the whole window (percentiles, shares of time, RMS) were used; sequence models or frequency-domain IMU features might capture short precursors that summaries miss. This was not tested.

Attrition. Only 37 of the 73 logs behind expert aircraft diagnoses with a checkable cause could still be downloaded, so several diagnostic comparisons in Table 7 have very small sample sizes ($n = 2$ to 9); its results are indicative rather than definitive.

Power. Some comparisons are small (14 pilot-caused matched sets, 12 sets with the newer vibration metric, three vibration-diagnosed logs). Null results there are not evidence of no effect.

Language models. The AI comparison is preliminary: the experiments use 60 author-generated cases and relatively small local models (3–4B parameters). The fixed rule's performance may be optimistic on cases created by the same authors who designed the rule (0 wrong answers is close to a training score); a blind set of cases written by someone who has not seen the rule is in preparation. Larger models or alternative prompt architectures were not tested.

The tool. AeroFleet 2.0's checks are primarily engineering tests on archive data and synthetic fault injections, not evidence of improved operational safety. Plans in the injection test were built from the flights themselves; the description search was tested on forum titles, which are shorter and more varied than an operator's report, and its symptom lists were written by the authors; prospective fleet evaluation of the vibration ranking and user studies of the three-dimensional views remain future work (Sections 6 and 7).

9. Conclusion

Across 479,547 public autopilot logs and 590 crash-rated flights, flight logs did not distinguish pilot-caused from aircraft-caused crashes above chance, configuration protections were disabled at similar rates, and the flight that crashed looked indistinguishable from the same drone's prior flights. Apparent diagnostic success was largely driven by crash-impact data leakage. What the logs did reveal, on older estimator metrics, was that drones that later crashed exhibited higher baseline vibration on ordinary flights than matched peers, though replication on newer accelerometer metrics was inconclusive and operational screening yields a low positive predictive value ($\approx 3.3\%$). Flight log analytics for drone fleets should prioritize longitudinal airframe screening and structured evidence review, not automated verdicts. AeroFleet 2.0 implements this approach as an open-source tool, presenting verifiable evidence while leaving diagnostic decisions to human operators.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The authors develop the open-source AeroFleet 2.0 tool described in Section 6; it is not commercial.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used an AI assistant (Claude, Anthropic) to write analysis code, draft and revise text, and assist the recording of forum diagnoses. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Data availability

Selection scripts, analysis code, the declared hypotheses with their commit history, derived features, expert labels (without forum text), and the source code of the AeroFleet 2.0 tool are available at `https://github.com/AdityaPathare46/aerofleet`; a DOI-archived release will be provided on acceptance. Raw logs are public on PX4 Flight Review and are referenced by identifier.

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Appendix A. Three-dimensional replay and mission rehearsal

Deviations, heights and separation between drones are spatial, and a two-dimensional plot hides height. The tool therefore draws flights on a virtual tabletop, a world in miniature [32], that can be viewed in a browser, in immersive virtual reality or, on a passthrough headset, in mixed reality on a real desk (WebXR).

A *flight replay* shows a reviewed flight: the approved plan dashed, the flown track green within tolerance and red during a deviation, the log's events pinned where they happened, and a playhead that can be scrubbed or played at ten times real speed.

A *mission rehearsal* shows planned missions before anyone flies: the deconfliction check turns each mission file into a position-over-time trajectory on a common clock (straight legs at stated climb, cruise and descent speeds), finds every pair closer than the separation distance and resolves conflicts by a launch delay or, where allowed, a different cruise height. The rehearsal flies all missions on that clock, as uploaded and as rescheduled, with each loss of separation drawn where and when it occurs (Figure A.7). It is a kinematic rehearsal of the plan, not a simulation of the aircraft: there is no wind, no vehicle dynamics and no sensor error, so it shows what the plan implies if it is flown as written, which is the assumption the planned-versus-flown check then tests after the flight.

The replay and rehearsal have been checked functionally on desktop; a paired standalone headset application compiles but has not been run on a

headset, and no user study has been carried out, so we make no claim that the three-dimensional view improves decisions.

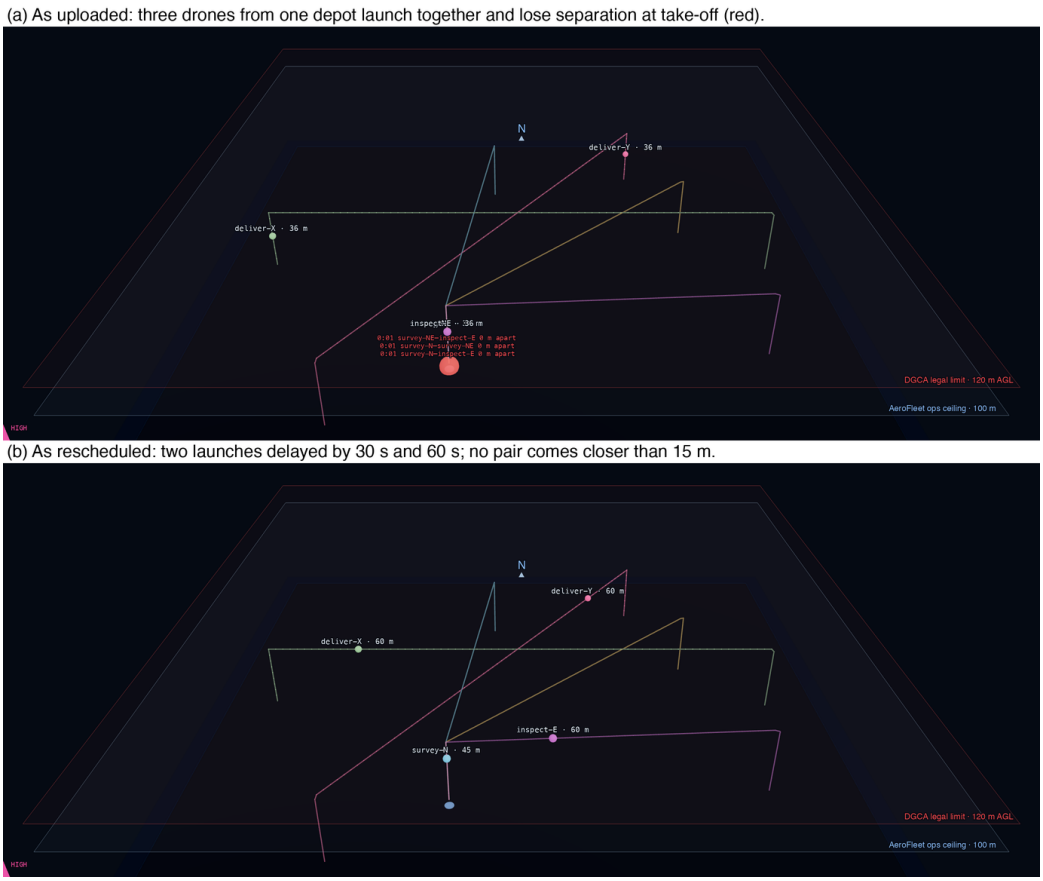


Figure A.7: Mission rehearsal in the tool's tabletop view (desktop rendering; the same scene opens in a WebXR headset). Five planned missions over a 2 km table, heights exaggerated four times, each mission in its own colour. (a) As uploaded, 12 s after launch: three missions leaving one depot together occupy the same point (overlapping labels) and the red sphere marks their loss of separation. (b) The schedule produced by the deconfliction check, 45 s after the first launch: two launches are delayed and no pair comes within 15 m. Straight legs at planned speeds, no wind.

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